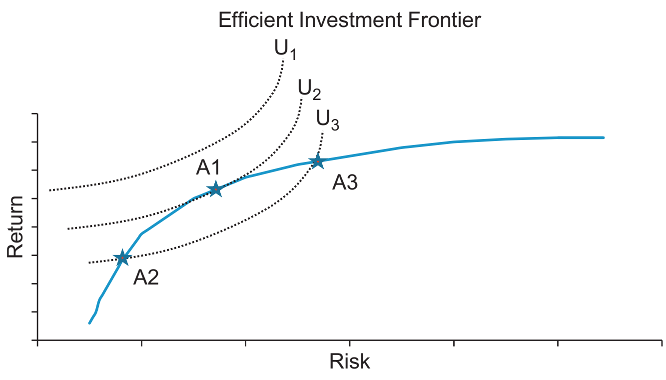
In Figure 10.2 we show the efficient frontier with three different utility curves. These indifference curves are ordered such that $U_1 > U_2 > U_3$. Therefore, investors will prefer portfolios on U_1 to portfolios on U_2 , and will prefer portfolios on U_2 to portfolios on U_3. Notice that these utility curves are shaped in such a way that investors will only accept more risk if they receive higher returns. For utility curve U_3 investors are equally happy with either portfolio A2 or A3 since they are on the same indifference curve. But investors prefer portfolio A1 to either A2 or A3 since utility curve U_2 is higher than utility curve U_3 ($U_2 > U_3$). Unfortunately, utility curve U_1 does not intersect with the efficient trading frontier and does not contain any optimal portfolios−it is an unattainable level. The best that investors can achieve is A1 on curve U_2.

This utility maximization proves to be an invaluable exercise for not only determining the preferred optimal portfolio, but also for determining the optimal trade schedule to achieve that optimal portfolio. We make further use of utility optimization below.

TRADING DECISION PROCESS

Once the optimal portfolio has been constructed traders are tasked with determining the appropriate implementation plan to acquire that new position. As discussed through the text, when implementing these decisions investors encounter the by now all too well known trader's dilemma—trading too quickly results in too much market impact cost but trading too slowly results in too much timing risk.

To determine the best way to implement the portfolio manager's decision, Almgren and Chriss (1999, 2000) provided a framework similar



■ **Figure 10.2** Maximizing Investor Utility.